

Meeting Notes

Contents

1	Meeting 1 [07/01/2026]	3
1.1	Agenda	3
1.2	Topics Discussed	3
1.3	Tasks	3
2	Meeting 2 [14/01/2026]	4
2.1	Agenda	4
2.2	Topics Discussed	4
2.3	Tasks	5
3	Meeting 3 [21/01/2026] - Cancelled	6
3.1	Agenda	6
3.2	Topics Discussed	6
3.3	Tasks	6
4	Meeting 4 [29/01/2026]	7
4.1	Agenda	7
4.2	Topics Discussed	7
4.3	Tasks	7
5	Meeting 5 [03/02/2026]	8
5.1	Agenda	8
5.2	Topics Discussed	8
5.3	Tasks	9

6 Meeting 6 [09/02/2026]	10
6.1 Agenda	10
6.2 Topics Discussed	10
6.3 Tasks	12
7 Meeting 7 [20/02/2026]	13
7.1 Agenda	13
7.2 Topics Discussed	13
7.3 Tasks	15

1 Meeting 1 [07/01/2026]

1.1 Agenda

- Validating the TER subject and obtaining an initial refocusing
- Knowing how to start the state of the art
- Choosing the tools to use
- Planning the next meeting

1.2 Topics Discussed

During the meeting, we validated the subject on partial preferences. The objective is really to explore this subfield of computational social choice theory by limiting ourselves to indivisible resources.

Afterwards, we listed some avenues for the state of the art. We will start by looking for articles on partial preferences focus on indivisible resources.

- Acceptability: Some researchers work on partial preferences under questions of acceptability by considering graphs (bipartite). For example, in the marriage problem, some agents refuse an allowance altogether. Thus, not all objects are allocable to a given agent. Acceptability therefore amounts to defining a list of preferences by excluding those on which the agent issues a right of veto.
- Additivity: Combining individual preferences into bundles of preferences adds information and complexity.
- Metric distortion problem: Ordinal preferences can be deduced from cardinal preferences and used as partial preferences. We then evaluate the extent to which the solution is degraded compared to what could have been proposed with the initial cardinal preferences.
- Cardinal preferences: Agents can express their preference in cardinal form, i.e. the intensity of the benefit they will derive from the allocation. Although this is suitable for evaluating equity criteria (maxmin, envy-free allocation), it is often difficult for agents to express such preferences.
- Truncated ballots: Derive from the vote theory.
- Uncertain preferences: The preferences are described by a probability distribution.

1.3 Tasks

For the next meeting on January 14, 2026, I will start the state of the art by listing partial/incomplete preference types.

2 Meeting 2 [14/01/2026]

2.1 Agenda

- Discussed on the partial preferences types found
- Clarifying some concepts
- Identifying key papers and resources
- Creating a GitHub repository for collaboration
- Planning the next meeting

2.2 Topics Discussed

During the meeting, we listed exhaustively the partial preferences. To prepare for the meeting, I had started reading the paper « Iterative voting with partial preferences » of Zoi Terzopoulou et al.

- Top k preferences: Agents only rank their top k choices, leaving the rest unranked.
- Truncated preferences: Similar to top k preferences, but the ranking is truncated at a specific point.
- Ordinal preferences: Agents express their preferences in a ranked order, without specifying the intensity of their preferences. It's a kind of distortion problem with respect to the cardinal preferences.
- Range of cardinal preferences: Agents provide a range of values for each option instead of a single value.
- Approval preferences: Agents indicate if a preference is approved or not, without ranking them. It's a binary indication and can be the result of boolean questions.
- Single peak preferences: Agents have a most preferred option, and their preference decreases as they move away from this peak (unimodal).
- Euclidean preferences: Agents' preferences are represented in a multi-dimensional space, where distance reflects preference intensity.
- Subset of possible orders: Agents provide a subset of possible rankings. The top k is a special case of this type.
- Combination of types: In practice, agents may express their preferences using a combination of the above types. For example, top k with structural aspects (single peak, euclidean).

Then, I learned how to search for articles on Google Scholar.

We also discussed researchers who have worked with partial preferences:

- Zoi Terzopoulou
- Sylvain Bouveret
- Jérôme Lang (director of LAMSADE): He studied the solutions that can occur and those that are valid (possible for a given problem).

Subsequently, we created a GitHub repository to collaborate on the TER.

Finally, we have also mentioned the value of partial information. In the problems considered, it is strongly assumed that agents clearly express their preferences. This work can be cognitively heavy for an agent who after a certain time will express himself at random. Then, sometimes the lack of information (for example on the ten candidates in an election) can prevent the expression of a complete ranking. How do you measure the value of a preference in a given context? This question is a matter of data elicitation and remains annexed to the study framework, although it justifies the importance of the study of partial preferences.

2.3 Tasks

For the next meeting on January 21, 2026, I will continue the state of the art by reading the articles.

3 Meeting 3 [21/01/2026] - Cancelled

3.1 Agenda

- Setting up the LaTeX environment
- Writing the notes of the previous meetings
- Creating a document to centralize my reading notes
- Planning the next meeting

3.2 Topics Discussed

To prepare for the meeting, I had started reading the paper "Fair Division under Ordinal Preferences: Computing Envy-Free Allocations of Indivisible Goods". This article studies ordinal preferences over indivisible goods and relates to concepts discussed during the previous meeting.

In particular, the authors derive partial preferences over bundles from the assumption that agents specify ordinal preferences over all individual goods. The rest of the paper studies fairness criteria, especially envy-freeness and Pareto efficiency, under this preference model.

During this week, I became familiar with the concepts of Necessary Envy-Freeness (NEF) and Possible Envy-Freeness (PEF), which involve strong properties for the solutions of the problem.

I also started paying closer attention to the mathematical notations used in the papers in order to better understand and formalize the concepts.

3.3 Tasks

For the next meeting on January 29, 2026, I will continue the state of the art by searching for articles on partial preferences. In parallel, I will learn how to write mathematical formulas in Latex.

4 Meeting 4 [29/01/2026]

4.1 Agenda

- Monitoring progress on the state of the art
- Writing the notes of the previous meetings and readings
- Planning the next meeting

4.2 Topics Discussed

To prepare for the meeting, I had started reading Chapter 10 of the Handbook of Computational Social Choice.

During the meeting, we discussed several points that I had highlighted, especially the following open questions:

- Coalitional manipulation problem: can it be related to a partial preference framework?
- Truly incomplete preferences: what does this mean? Is it an elicitation problem or a problem of uncertainty? This notion was studied by Pini et al. (2009) and Xia and Conitzer (2011b).

The first question refers to Chapter 2 of the Handbook of Computational Social Choice, which studies coalition formation. More precisely, agents can be grouped into coalitions with shared or compatible preferences. This allows a more compact representation of preferences. However, this perspective goes beyond the current scope of our study.

This led to a broader discussion on hedonic games, where agents form coalitions and have preferences within the coalitions they belong to. This can be viewed as a restriction on the set of possible preferences. We noted that this connects to our broader reflection on partial preferences and structured preference domains. We also discussed the possibility of contributing to related questions later in the project.

In our study, once a clear research question is identified, we will formulate assumptions on the structure of the available information, without questioning us on the elicitation process itself. Hedonic games provide an example of structured or restricted preference domains that can be interpreted as a form of partial preference information. In addition, agents may classify other agents into categories such as friends, enemies, or unknown agents.

The second question requires further exploration.

We also discussed preference distribution models, including Impartial Culture and the Polya-Eggenberger urn model. Under Impartial Culture, each ranking is equally likely to represent an agent's preferences and is independent from each other. In contrast, in the Polya-Eggenberger urn model, the probability of a ranking being drawn is proportional to the number of times it has already been drawn. More precisely, the urn model is a random draw with replacement of α copies. This model allows us to capture the idea that some rankings are more likely than others, which is often the case in real-world scenarios. It describes correlated preferences.

4.3 Tasks

For the next meeting on February 3, 2026, I will continue the state of the art by studying the survey by Jérôme Lang and continuing the reading of Chapter 10 of the Handbook of Computational Social Choice. The objective is to identify potential gaps in the literature that could lead to a research question.

5 Meeting 5 [03/02/2026]

5.1 Agenda

- Discussed following the reading of Jérôme Lang's survey and the chapter 10 of the Handbook of Computational Social Choice
- Updating the list of partial preferences
- Understanding the cross-domain aspect of "possible" and "necessary" properties.
- Identifying key papers and resources
- Planning the next meeting

5.2 Topics Discussed

To prepare for the meeting, I had started reading the paper the chapter 10 of the Handbook of Computational Social Choice and the Jérôme Lang's survey "Collective Decision Making under Incomplete Knowledge: Possible and Necessary Solutions" (2020).

Updated list of partial preferences:

1. **Top k preferences:** Agents only rank their top k choices, leaving the rest unranked.
2. **Bottom-k preferences:** Agents only rank their bottom k alternatives, leaving the rest unranked.
3. **Truncated preferences:** Similar to top k preferences, but the ranking is truncated at a specific point. Truncated ballots.
4. **Ordinal preferences:** Agents express their preferences in a ranked order, without specifying the intensity of their preferences. It's a kind of distortion problem with respect to the cardinal preferences.
 - Ordinal preferences over the set of single goods, induced partial strict linear order over bundles.
5. **Range/Interval of cardinal preferences:** Agents provide a range of values for each option instead of a single value.
6. **Approval preferences:** Agents indicate if a preference is approved or not, without ranking them. It's a binary indication and can be the result of boolean questions. Agents approve a fixed number of alternatives or any number of alternatives.
7. **Veto preferences:** Agents disapprove a fixed number of alternatives or any number of alternatives. The k-veto is equivalent to (m-k)-approval preferences with m the number of alternatives.
8. **Full ranking preferences:** Agents provide a complete ranking of all alternatives.
9. **Next-best alternative queries:** Agent provide in sequential order her most preferred alternative among those not yet ranked.
10. **Top-k queries:** Agents approve k alternatives and give no order between them. Agents answer sequentially to the question "What are your most preferred alternatives?"
11. **Positional queries:** ask agent to provide the alternative she ranks at a given position t.
12. **Pairwise comparisons:** Agents indicate which of the both options considered is preferred.

13. **Single peak preferences:** Agents have a most preferred option, and their preference decreases as they move away from this peak (unimodal).
14. **Euclidean preferences:** Agents' preferences are represented in a multi-dimensional space, where distance reflects preference intensity.
15. **Subset of possible orders:** Agents provide a subset of possible rankings. The top k is a special case of this type.
16. **Choice queries:** Agents indicate which alternative in a subset S is most preferred. Kind of clustering approach.
17. **Set ranking:** Agents provide a ranking in a subset of alternatives.
18. **Split queries:** ask agent to partition a subset of alternatives into two or more subsets and order them such that all alternatives in a higher-ranked subset are preferred to all alternatives in a lower-ranked subset. k-approval is a special case where two blocks of size k and m-k are created.
19. **Combinaison of types:** In practice, agents may express their preferences using a combination of the above types. For example, top k with structural aspects (single peak, euclidean).
20. **Incomplete Set of preference profiles:** Only the preference profile of a subset of agents is known.
 - Application: The votes of nonmanipulators are fully specified, while no information is given for the manipulators.
21. **Truly incomplete preferences:** ???
22. **Probabilistic preferences:** Agents' preference profile follow a probabilistic distribution. Maximum Likelihood Estimator to estimate the distribution characteristics from a sample of preferences. An alternative can be more likely than others.
23. **Attribute preference:** Agents have different preference profile depending on the attribute considered (politics, economics, immigration).

We also discussed a few points:

- Considering only the possible winners and exclude the losers in certain situation without affecting the problem solving.
- Binary axis to represent political affiliation may be simplistic. Unimodal can be limited to describe the reality.
- Multiwinner: avoiding this topic for our state of the art. It is a field in extension.

5.3 Tasks

For the next meeting on February 9, 2026, I will continue the state of the art by searching for articles on partial preferences expressed as intervals. I will also prepare the discussion by reading some articles: Kenig et Kimelfeld (2019) on MLE; Pini et al. (2009); Xia and Conitzer (2011b). The objective is to determine whether it is interesting to investigate deeper in these topics and extract research questions. I would also like to write my notes neatly.

6 Meeting 6 [09/02/2026]

6.1 Agenda

- Discussion of recent readings
- Planning the next step and future meeting

6.2 Topics Discussed

To prepare for the meeting, I conducted a brainstorming session to identify key concepts, reflections, and questions arising from my recent readings and from previous discussions. The following points were noted:

- Agents can be partially known.
- Some voting rules cannot be defined solely as a function of a voter's ranking. This is the case for approval voting and range voting, which require additional information beyond a strict ordinal ranking.
- Mallows Ranking Models: probabilistic frameworks for modeling and analyzing rankings.
- Complete preference information is not always necessary to reach a decision. A key question is determining how much information, or which structural properties, are sufficient to conclude. This motivates the study of *necessary* and *possible* outcomes. Instead of considering all allocations, one can restrict attention to possible allocations, and further to necessary allocations, thereby reducing the search space by excluding impossible solutions.
- It is sometimes possible to translate preferences from one representation to another. For example, ordinal preferences can be derived from cardinal preferences, although this transformation results with a loss of information.
- The notion of an interval representation requires clarification. In particular, it is important to understand how intervals relate to linear orders, and whether an interval representation such as $[a, b, c]$ can be interpreted as a weak order (e.g., $a \succeq b \succeq c$) or whether it conveys fundamentally different information.

I then continued the state-of-the-art review by skimming the article by Kenig and Kimelfeld (2019), as well as the article "Representing Preferences Using Intervals" by Öztürk, Pirlot, and Tsoukiàs (2011).

The first article studies positional scoring rules under probabilistic voting models, in particular when preferences are incomplete. The authors consider two probabilistic models: a uniform distribution over all completions of a partial voting profile, and the Repeated Insertion Model (RIM). Each voter is associated with a scoring vector that assigns points to candidates depending on their position in the ranking.

- **Plurality rule:** the vector contains a 1 for the most preferred alternative and 0 for all others.
- **Borda rule:** the vector contains $m - 1$ points for the most preferred alternative, $m - 2$ for the second most preferred alternative, and so on down to 0 for the least preferred alternative. Candidate receives points depending on its rank in each voter's preference list, and the candidate with the highest total score wins.
- **k-approval rule:** the vector contains k entries equal to 1 for the top k the approved alternatives and $m - k$ entries equal to 0 for the disapproved alternatives.

Using these scoring vectors, the authors analyze the computational complexity of estimating the probability of an election outcome under uncertainty. Their analysis relies on structural properties of voting rules, in particular the notions of *pure* and *almost constant* scoring rules.

A voting rule is *pure* if the scoring vector for m candidates can be obtained from the scoring vector for $m - 1$ candidates by inserting a score for the new candidate without affecting the score of the other candidates. All the previously mentioned rules satisfy this property.

A voting rule is *almost constant* if its scoring vector contains $m - K$ identical scores for some fixed constant K . In other words, only a fixed number of candidates receive distinct scores, while the remaining candidates share the same score. The Borda rule does not satisfy this property.

An open question discussed during the meeting was whether the notion of almost constant scoring rules can be extended to preference representations based on intervals.

More broadly, the article expresses the incompleteness/partial preferences and the probability as subfields of the uncertainty.

Memo: Do not confuse partial profile with partial vote.

- **Partial profile:** all partial votes of voters.
- **Partial vote:** partial preferences of a single voter.

The second article provided a different perspective on preference representation using semiorders (i.e interval orders). Several observations were noted:

- **Luce's cup of coffee:** small differences in utility may not be distinguishable, even though larger differences are meaningful. This motivates interval-based representations of preferences. More illustratively, there is no difference between a little more sugar and no sugar at all. However, it is preferable to have more sugar than not at all to soften the bitter taste of coffee.
- **Transitivity:** indifference is not necessarily transitive in semiorders. An agent may be indifferent between a and b , and between b and c , without being indifferent between a and c . This is an important consideration when working with interval representations.
- **Fishburn's classification:** Fishburn identifies nine types of semiorders, providing a theoretical framework for studying such representations.

However, during the meeting, we decided not to include this article in the state of the art, as its definition of intervals differs from the one relevant to our project. In our work, intervals are considered in the context of responsive set extensions.

To illustrate the concept of responsive set extension, we consider a set of objects with cardinal utilities:

objects	o_1	o_2	o_3	o_4
cardinal preferences	10	9	8	7
strict ordinal preference	$o_1 \succ o_2 \succ o_3 \succ o_4$			

From this ranking, we can conclude, for example, that: $\{o_1, o_2\} \succ \{o_3, o_4\}$

However, some bundle comparisons remain undetermined, such as between $\{o_1, o_4\}$ and other bundles.

The objective is to give a formal interpretation to the following representation:

objects	o_1	o_2	o_3	o_4
intervals	$[5, 8]$	$[6, 7]$	$[4, 7]$	$[8, 10]$
deduced ordinal preferences	5	7	6	8

Using these intervals, the value of bundles may be obtained by summing interval bounds. For example:

$$v(\{o_1, o_2\}) \in [5 + 6, 8 + 7] = [11, 15]$$

$$v(\{o_1, o_3\}) \in [5 + 4, 8 + 7] = [9, 15]$$

$$v(\{o_3, o_4\}) \in [4 + 8, 7 + 10] = [12, 17]$$

However, not all values within an interval are equally plausible, as some combinations of individual utilities may be more likely than others. For instance, the values $6+6 = 12$ and $5+7 = 12$ for the bundle $\{o_1, o_2\}$.

From this representation, the evaluation of possible/necessary envy-freeness or other properties is enabled.

6.3 Tasks

For the next meeting on February 20, 2026, I will continue the state-of-the-art review by studying responsive set extensions, in particular the articles by Aziz et al. (2024) and Barberà et al. (2001).

If time permits, I will also explore envy-freeness properties for partial preferences under other contexts.

7 Meeting 7 [20/02/2026]

7.1 Agenda

- Discussion of recent readings on responsive set extensions and envy-freeness
- Identification of potential research directions
- Planning the next meeting

7.2 Topics Discussed

To prepare for the meeting, I began reading the article “Computational Complexity of Necessary Envy-Freeness” by Aziz et al. (2024).

Before discussing the article in detail, we reviewed the main research questions that have already been addressed in the literature and that are related to our topic.

Existence and complexity of envy-freeness. The problem of checking whether there exists a complete necessary envy-free (NEF) allocation is NP-complete in general, and polynomial-time solvable when there are only two agents. In contrast, checking the existence of a complete possible envy-free (PEF) allocation is computationally tractable. Related work also studies envy-freeness for bundles induced by ordinal preferences (“Fair Division under Ordinal Preferences: Computing Envy-Free Allocations of Indivisible Goods” by Bouveret et al.).

Possible and necessary winner problems. Given a partial voting profile and a positional scoring rule:

- The *possible winner problem* asks whether a given candidate can win in at least one completion of the profile. This problem is NP-complete in general, but polynomial-time solvable for specific rules such as plurality and veto.
- The *necessary winner problem* asks whether a candidate wins in every completion of the partial profile. This problem is polynomial-time solvable for all positional scoring rules.

Winning probability problem. Another related line of work studies the computational complexity of estimating the probability of an election outcome under uncertainty (e.g., Kenig and Kimelfeld, 2019).

Inference rules for bundle preferences. Several works analyze how preferences over bundles can be inferred from preferences over single goods. In particular, inference rules based on linear orders over single goods have been studied (e.g., Brams and King; Brams et al.).

At this stage, we have sufficient material to write a structured state of the art. The focus of the meeting shifted toward identifying a clear research question.

Discussion on Aziz et al. (2024)

During the meeting, we discussed the approach taken by Aziz et al. to describe the envy freeness properties. More precisely, an agent is said to prefer one bundle to another if each item in the second bundle can be

matched to a distinct item in the first bundle that the agent strictly prefers. In other words, the agent can pair every item in the less preferred bundle with a strictly better item in the other bundle.

The extension of preferences induced partial preferences, which should be included in our state of the art.

Clones and Structured Preferences

We then discussed the notion of *clones*, as mentioned in the survey by Jérôme Lang. Two alternatives are considered clones if they always appear consecutively in preference rankings (although possibly in different internal orders across agents). Importantly, no external alternative can appear between them.

For instance, in a political election context, two ideologically close candidates may behave as clones: they always appear consecutively in voters' rankings, although their internal order may vary across voters.

The notion of clones is particularly interesting for our project because it suggests a structural form of indeterminacy that is conceptually close to interval-based representations.

This led to the following perspective:

Suppose that objects can be partitioned into categories (or groups), where objects within the same group are substitutes. For example, train and plane tickets to the same destination may be substitutable, whereas a train ticket and a hotel booking may be complementary. If we allocate together the train ticket and the hotel booking, the bundle will have more impact than if we allocate the train ticket with the plane ticket.

Hence, we consider the following structured model:

- Agents have strict linear preferences over single objects.
- The set of objects is partitioned into k groups $M_1, \dots, M_k$ with $k \geq 1$.
- Objects within each group M_j are clones: they appear consecutively in each agent's ranking, though possibly in different internal orders.
- The ranking between groups is known.
- The internal ranking within each group is unknown.

This framework can be interpreted as a structured form of partial preferences. It generalizes the baseline case (ordinal preference profile) in which each group contains exactly one object.

An open question is whether this model corresponds to an already studied form of partial preference representation.

Resulting questions.

- Under clone-group structured partial preferences, what is the complexity of checking existence of NEF allocation?
- Does clone-group structure make the NEF problem tractable?
- Does clone-group uncertainty behave similarly to matching uncertainty?

Connections with Matching Theory

We briefly discussed potential links with matching theory.

Stable matching. In the stable marriage setting, instability arises when a student and a university prefer each other to their assigned partners.

Popular matching. A matching A is popular if there is not exist another matching A' such as the number of agents who prefer the matching A' is superior to the number of agents who prefer the matching A . A popular matching does not always exist, and its existence must be characterized.

The matching is a bijection between the set of agents and the set of objects. In our model, the set of object is divisible into $|N|$ groups (a group can be empty).

Matching problems may provide useful tools for analyzing allocations under structured partial preferences.

7.3 Tasks

For the next meeting on March 9, 2026:

- Study the literature on popular matchings.
- Investigate the notion of cloning in more detail, starting from the relevant section in Jérôme Lang's survey.
- Upload structured notes and begin drafting the state of the art.